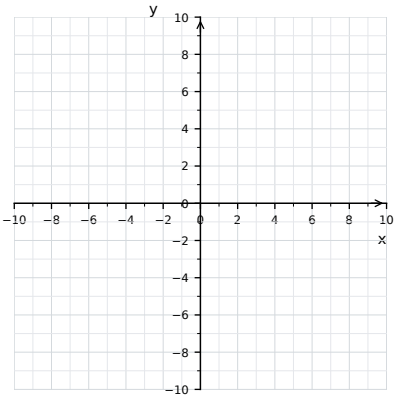
TSM 099 Final Exam

Practice 1 | Version 202610122020.31

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right.

QUESTIONS	ANSWERS
1. Find the slope-intercept equation through these two points. $(2,\ -1);\ (4,\ 0)$	1. Equation 10 pts
2. Factor completely. $6x^2 - 5x - 6$	2. Factors 10 pts
3. Find the intercepts and graph the line on the grid. $2x + y = 6$	3. Intercepts 4 pts Graph  6 pts
4. Solve by taking square roots. $3(x-1)^2 = 27$	4. Solutions 10 pts
5. Add and simplify. $\frac{2}{x^2-2x} + \frac{3}{x^2-4}$	5. Expression 10 pts
6. Solve using the quadratic formula. Give exact simplified solutions. $2x^2 = 4 - x$	6. Solutions 10 pts
7. Solve by factoring. $x^2 + 2x = 8$	7. Solutions 10 pts
8. Solve and check in the original equation. $\frac{8}{2x-6} + \frac{2}{x-3} = 2$	8. Solution/check 10 pts

TSM 099 Final Exam

Practice 1 | Version 202610122020.31

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right.

QUESTIONS	ANSWERS
9. Multiply and simplify. $\frac{x^2-4}{x^2-2x-35} \times \frac{x+5}{x^2+2x}$	9. Expression 10 pts
10. Evaluate at x = -2; show substitution. $(x+3)(2x-2)$	10. Value 10 pts

Instructor answer key: TSM 099 Final Exam

Practice 1 | Version 202610122020.31

Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. Equation

$y = \frac{x}{2} - 2$

10 pts

2. Factors

$(2x - 3)(3x + 2)$

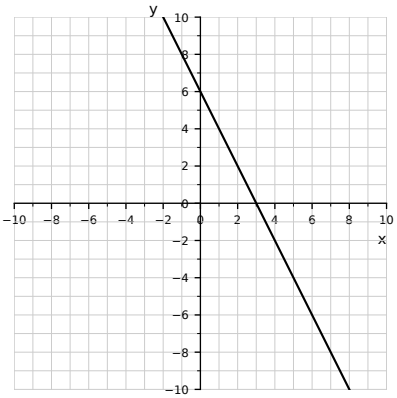
10 pts

3. Intercepts

$(3,\ 0);\ (0,\ 6)$

4 pts

Graph



6 pts

4. Solutions

$x = -2,\ 4$

10 pts

5. Expression

$\frac{5x + 4}{x(x - 2)(x + 2)}$

10 pts

6. Solutions

$x = -\frac{1}{4} + \frac{\sqrt{33}}{4},\ \ -\frac{\sqrt{33}}{4} - \frac{1}{4}$

10 pts

7. Solutions

$x = -4,\ 2$

10 pts

8. Solution/check

$x = 6$

10 pts

TEST PAGE 2

9. Expression

$\frac{x - 2}{x(x - 7)}$

10 pts

10. Value

-6

10 pts